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SERVER CANISTER FOR LIQUID-COOLED SERVER RACKS

Abstract

A canister for coupling a plurality of servers to a server rack is described herein. The canister includes an influent rigid manifold and an effluent rigid manifold that have respective pluralities of quick-disconnect (QD) fittings arranged in fixed orientations and configured to couple with corresponding QD fittings on the servers. The canister further includes an influent canister QD fitting coupled with the influent rigid manifold and configured to couple with an influent soft manifold of the server rack. The canister also includes an effluent canister QD fitting coupled with the effluent rigid manifold and configured to couple with an effluent soft manifold of the server rack. The canister is one of many canisters configured to mount to the server rack. By using multiple canisters and the soft manifolds, many liquid-cooled servers may be implemented within the server rack without necessitating high tolerance manufacturing of the manifolds.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/561,222 filed on Mar. 4, 2024. The entire disclosure of U.S. Provisional Application No. 63/561,222 is incorporate by this reference.

FIELD
[0002] This disclosure is directed to liquid cooling of servers within a server rack.
BACKGROUND
[0003] Liquid cooling is quickly becoming the norm for server systems (e.g., artificial intelligence (AI), machine learning (ML), and high-performance computing (HPC) systems). Server racks often contain tens of servers (e.g., 42 1U servers in a 42u rack). To cool the servers within the server racks, the server racks often contain rigid liquid manifolds with a plurality of quick disconnect (QD) fittings mounted in fixed orientations. As the number of liquid-cooled servers (and, thus, QD fittings) increases, so does the tolerance stack up within the rigid liquid manifolds (especially in the height direction). The tolerance stack up can make the necessary connections with the servers difficult to make.
SUMMARY
[0004] A canister for coupling a plurality of servers to a server rack is described herein. The canister includes an influent rigid manifold that has a plurality of influent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with influent server QD fittings on the servers. The canister also includes an effluent rigid manifold that has a plurality of effluent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with effluent server QD fittings on the servers. The canister further includes an influent canister QD fitting coupled with the influent rigid manifold and configured to couple with an influent soft manifold QD fitting of an influent soft manifold of the server rack. The canister also includes an effluent canister QD fitting coupled with the effluent rigid manifold and configured to couple with an effluent soft manifold QD fitting of an effluent soft manifold of the server rack.
[0005] A cooling system for a server rack is also described herein. The cooling system includes an influent soft manifold having a plurality of influent hoses and a plurality of influent soft manifold QD fittings at respective ends of the influent hoses. The cooling system also includes an effluent soft manifold having a plurality of effluent hoses and a plurality of effluent soft manifold QD fittings at respective ends of the effluent hoses. The cooling system further includes a plurality of the canisters described above.
[0006] A server rack is also described herein. The server rack includes a server rack frame and the cooling system described above. The canisters are configured to mount to the server rack frame.
[0007] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description. In the drawings, like reference numbers indicate identical or functionally similar elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** illustrates an example of a server rack with a cooling system including canisters in accordance with this disclosure.

[0009] FIG. **2** illustrates an example of an influent soft manifold in accordance with this disclosure.

[0010] FIG. **3** illustrates an example of a canister mounted to a server frame in accordance with this disclosure.

[0011] FIG. **4** illustrates an example of a configuration of canisters within a rack in accordance with this disclosure.

[0012] FIG. **5** illustrates an example of an influent rigid manifold, influent canister hose, and influent canister QD fitting in accordance with this disclosure.

[0013] FIG. **6** illustrates an example of mounting a canister to a server frame in accordance with this disclosure.

[0014] FIG. **7** illustrates an example of mounting a server to a canister that is mounted to a server frame in accordance with this disclosure.

DETAILED DESCRIPTION

Overview

[0015] Liquid cooling of servers within a server rack is often facilitated via rigid manifolds that are made of welded stainless-steel tubes with QD fittings attached to branches of the manifolds. It can be difficult to manufacture such manifolds for use with server racks configured to accept large numbers of servers. For example, tolerance stack-up in a vertical direction can cause fluid connections to be difficult to make and/or to be unreliable (e.g., leak).

[0016] Described herein is a canister for coupling a plurality of servers to a server rack. The canister includes an influent rigid manifold and an effluent rigid manifold that have respective pluralities of quick-disconnect (QD) fittings arranged in fixed orientations and configured to couple with corresponding QD fittings on the servers. The canister further includes an influent canister QD fitting coupled with the influent rigid manifold and configured to couple with an influent soft manifold of the server rack. The canister also includes an effluent canister QD fitting coupled with the effluent rigid manifold and configured to couple with an effluent soft manifold of the server rack. The canister is one of many canisters configured to mount to the server rack.

[0017] By using multiple canisters and the soft manifolds, many liquid-cooled servers may be implemented within the server rack without requiring high tolerance manufacturing. Tolerance stack-up can be mitigated because each of the rigid manifolds only services a subset of the servers of the rack.

[0018] In the following description, numerous specific details are set forth, such as particular structures, components, materials, dimensions, processing steps and techniques, in order to provide an understanding of the various embodiments of the present application. However, it will be appreciated by one of ordinary skill in the art that the various embodiments of the present application may be practiced without these specific details. In other instances, well-known structures or processing steps have not been described in detail in order to avoid obscuring the present application.

Example Server Rack

[0019] FIG. **1** illustrates an example of a server rack **100** incorporating a cooling system **102** utilizing canisters **104**. The cooling system **102** may have various components thereof attached to a server rack frame **106**. The server rack frame **106** may be a conventional server rack. Thus, the server rack **100** may comprise a conventional server rack with the cooling system **102** attached thereto.

[0020] In the illustrated example, the cooling system **102** contains four of the canisters **104** that are similar to each other. For example, the cooling system **102** may be configured to interface with 12 servers with each canister **104** configured to accept three of them. It should be noted that a three-server rigid manifold is much easier to manufacture within tolerance than a 12 server one. The canisters **104** are configured to couple respective pluralities of servers to the server rack frame **106**. For example, each canister **104** may be configured to couple three or more servers to the server rack frame **106**.

[0021] The cooling system **102** may contain any configuration and number of canisters **104** without departing from the scope of this disclosure. For example, the canisters **104** may all be similar (e.g., configured to couple a same number/size of server) or one or more of the canisters **104** may be different from the others (e.g., configured to couple a different number/size of server). Each of the canisters **104** may be configured to accept either 19-inch or 21-inch servers, for example.

[0022] Feeding the canisters **104** is an influent soft manifold **108**. The influent soft manifold **108** is configured to receive cooling liquid from a source (e.g., infrastructure cooling system), split the cooling liquid into a plurality of influent flows (e.g., via branches), and provide the influent flows to the canisters **104**. The influent soft manifold **108** may be coupled with a left side **110** of a soft manifold frame **112**. The soft manifold frame **112** may be attached to the server rack frame **106** or be a part of the server rack frame **106**.

[0023] Receiving the cooling liquid from the canisters **104** is an effluent soft manifold **114**. The effluent soft manifold **114** is configured to receive effluent flows from the canisters **104** (e.g., after passing through the servers), combine the effluent flows, and provide the cooling fluid to a sink (e.g., the infrastructure cooling system). The effluent soft manifold **114** may be coupled with a right side **116** of the soft manifold frame **112**. The soft manifold frame **112** may include a top **118** and/or a bottom **120** connecting the left side **110** and the right side **116**.

[0024] The orientation of the influent and effluent sides may vary without departing from the scope of this disclosure. For example, the influent soft manifold **108** and the effluent soft manifold **114** may be switched (e.g., cooling fluid goes from right to left instead of left to right). Furthermore, although the connections to the influent and effluent soft manifolds are on top of the server **100**, either or both of them may come out one of the sides or the bottom of the server **100**.

Example Influent Soft Manifold

[0025] FIG. **2** illustrates an example of the influent soft manifold **108** coupled with the left side **110** of the soft manifold frame **112**. To access and/or service the influent soft manifold **108**, the top **118** and the bottom **120** of the soft manifold frame **112** may be removed such that the influent soft manifold **108** and the left side **110** may be removed from the server rack **100**. The effluent soft manifold **114** may be removed similarly.

[0026] The influent soft manifold **108** contains an influent feeder QD fitting **200** that is configured to attach to a cooling liquid supply (e.g., from an infrastructure cooling system). Attached to the influent feeder QD fitting **200** is an influent feeder **202**. The influent feeder **202** is configured to supply a manifold portion of the influent soft manifold **108** with the cooling liquid. The influent feeder **202** is coupled with a first of a plurality of influent branch fittings **204**. The influent branch fittings **204** are configured to supply the cooling liquid to a plurality of influent hoses **206**. For example, the influent branch fittings **204** may be tees (pipe or tubing tees). In some implementations, an influent branch fitting **204** furthest from the influent feeder **202** may be an ell. In other implementations, the influent hoses **206** may all be coupled with the influent feeder **202** (e.g., in a multiple (four in the illustrated example)-to-one fitting). At the ends of the influent hoses **206** are influent soft manifold quick-disconnect (QD) fittings **208**. The influent soft manifold QD fittings **208** are configured to connect with influent canister QD fittings on the canisters **104**.

[0027] QD fittings as described herein may be any type of fluid connector. A QD fitting may be a simple hose connection (e.g., barb fitting or tubing connector) or a mating connector (e.g., blind mate, push-to-connect, twist lock, or sanitary). The types of QD fittings may vary within the system (e.g., different types for the various connections) without departing from the scope of this disclosure.

[0028] The influent feeder **202** and fluid connections between the influent branch fittings **204** may be soft (e.g., flexible) or rigid. The influent hoses **206**, however, are configured to be soft such that they can move relative to the rest of the influent soft manifold **108**. Soft in the context of this disclosure means the ability to deform, at least slightly, without undue force and/or without breaking. For example, the influent hoses **206** may be rubber or flexible plastic hose or tubing. The rest of the influent soft manifold **108** may be formed of rigid pipe or tubing (e.g., metal or plastic), flexible hose or tubing, or some combination thereof. In some implementations, the influent hoses **206** may be attached to a rigid manifold (e.g., a rigid structure of tubing or piping) of the influent soft manifold **108**. Regardless of how it is constructed, the influent soft manifold **108** is configured to allow the influent soft manifold QD fittings **208** to move relative to one another.

Example Canister

[0029] FIG. **3** illustrates an example of the canister **104**. Specifically, FIG. **3** illustrates a rear of the canister **104** (e.g., configured to be towards a rear of the server rack **100**) including fluid connections thereon. The canister **104** may include a canister frame formed of sheet metal. The canister **104** may contain a rear wall **300** to which various components may be mounted. The rear wall **300** may be intermittent (e.g., have two portions, a left portion and a right portion, or jog). The rear wall **300** may be configured to straddle a center member of the server rack **100** (as shown).

[0030] The canister **104** may contain an influent canister QD fitting **302** that is configured to couple with an influent soft manifold QD fitting **208**. The influent canister QD fitting **302** may be mounted to the canister **104** (e.g., the rear wall **300**) via an influent canister QD bracket **304**. The influent canister QD bracket **304** may facilitate easy connection between the influent QD fittings. Coupled with the influent canister QD fitting **302** may be an influent canister hose **306**. The influent canister hose **306** may be soft or rigid and is configured to couple the influent canister QD fitting **302** to an influent rigid manifold of the canister **104**.

[0031] The canister **104** may also contain an effluent canister QD fitting **308** that is configured to couple with an effluent soft manifold QD fitting (shown in FIG. **4**). The effluent canister QD fitting **308** may be mounted to the canister **104** (e.g., the rear wall **300**) via an effluent canister QD bracket **310**. The effluent canister QD bracket **310** may facilitate easy connection between the effluent QD fittings. Coupled with the effluent canister QD fitting **308** may be an effluent canister hose **312**. The effluent canister hose **312** may be soft or rigid and is configured to couple the effluent canister QD fitting **308** to an effluent rigid manifold of the canister **104**.

[0032] In some implementations, the influent soft manifold **108** may be directly coupled with the canister **104** (e.g., an influent hose **206** may be directly coupled with the influent rigid manifold), and/or the effluent soft manifold **114** may be directly coupled with the canister **104** (e.g., an effluent hose of the effluent soft manifold **114** may be directly coupled with the effluent rigid manifold).

Example Effluent Connections

[0033] FIG. **4** illustrates an example of four canisters **104** connected to the effluent soft manifold **114**. The effluent soft manifold **114** is shown coupled with the right side **116** of the soft manifold frame **112**. To access and/or service the effluent soft manifold **114**, the top **118** and the bottom **120** of the soft manifold frame **112** may be removed such that the effluent soft manifold **114** and the right side **116** may be removed from the server rack **100**. In the illustrated example, the effluent soft manifold **114** is connected to the canisters **104**.

[0034] The effluent soft manifold **114** contains an effluent return QD fitting **400** that is configured to attach to a cooling liquid return (e.g., of an infrastructure cooling system). Attached to the effluent return QD fitting **400** is an effluent return **402**. The effluent return **402** is configured to collect the cooling liquid from a manifold portion of the effluent soft manifold **114**. The effluent return **402** may be coupled with a first of a plurality of effluent branch fittings **404**. The effluent branch fittings **404** may be configured to gather the cooling liquid from a plurality of effluent hoses **406**. For example, the effluent branch fittings **404** may be tees (pipe or tubing tees). In some implementations, an effluent branch fitting **404** furthest from the effluent return **402** may be an ell.

In other implementations, the effluent hoses **406** may all be coupled with the effluent return **402** (e.g., in a multiple (four in the illustrated example)-to-one fitting). At the ends of the effluent hoses **406** are effluent soft manifold QD fittings **408**. The effluent soft manifold QD fittings **408** are configured to connect with the effluent canister QD fittings **308** on the canisters **104**.

[0035] The effluent return **402** and fluid connections between the effluent branch fittings **404** may be soft (e.g., flexible) or rigid. The effluent hoses **406**, however, are configured to be soft such that they can move relative to the rest of the effluent soft manifold **114**. Soft in the context of this disclosure means the ability to deform, at least slightly, without undue force and/or without breaking. For example, the effluent hoses **406** may be rubber or flexible plastic hose or tubing. The rest of the effluent soft manifold **114** may be formed of rigid pipe or tubing (e.g., metal or plastic), flexible hose or tubing, or some combination thereof. In some implementations, the effluent hoses **406** may be attached to a rigid manifold (e.g., a rigid structure of tubing or piping) of the effluent soft manifold **114**. Regardless of how it is constructed, the effluent soft manifold **114** is configured to allow the effluent soft manifold QD fittings **408** to move relative to one another.

[0036] As discussed above, the effluent canister hoses **312** are coupled with effluent rigid manifolds **410**. The effluent rigid manifolds **410** are attached to the canisters **104** and have a plurality of effluent rigid manifold QD fittings **412** arranged in a fixed orientation. The effluent rigid manifolds **410** may be configurable (relative to the canister **104** and/or relative to itself) for different sized servers.

Example Rigid Manifold

[0037] FIG. **5** illustrates an example of an influent rigid manifold **500** and associated components. The effluent rigid manifold **410** (and its associated components) may be configured similarly. The influent rigid manifold **500** may be formed of a single piece (e.g., square tubing) or be a plurality of pieces connected together. For example, the influent rigid manifold **500** may have one or more rotatable portions usable to configure the influent rigid manifold **500** for use with different servers. The influent rigid manifold **500** is rigid, at least in a vertical (e.g., lengthwise) direction, such that a vertical spacing between ports remains constant.

[0038] Attached to the influent rigid manifold **500** is a plurality of influent rigid manifold quick-disconnect (QD) fittings **502** (e.g., blind mate fittings). The influent rigid manifold QD fittings **502** are arranged in a fixed configuration (e.g., they are fixed relative to each other in a certain configuration of the influent rigid manifold **500**). For example, the influent rigid manifold QD fittings **502** may be arranged in a linear array. In some implementations, the influent rigid manifold **500** may be formed of welded tubes with the influent rigid manifold QD fittings **502** attached to tubing branches of the influent rigid manifold **500**.

[0039] Coupled with the influent rigid manifold **500** is the influent canister QD fitting **302**. The influent canister QD fitting **302** may be coupled to the influent rigid manifold **500** via the influent canister hose **306**. In some implementations, the influent canister QD fitting **302** may be rigidly coupled with the influent rigid manifold **500**. The influent rigid manifold **500** may also have mounting portions **504** (e.g., threaded holes or rivet nuts) that are configured to facilitate mounting the influent rigid manifold **500** to the canister **104**.

[0040] The influent rigid manifold **500** (as well as the effluent rigid manifold **410**) may be disposed on either side of the rear wall **300**. That is, in some implementations, the assembly may be mounted in an interior side of the canister **104** with the influent canister hose **306** (and the effluent canister hose **312**) going through the rear wall **300**. In other implementations, the assembly may be mounted on an exterior side of the canister **104** with the influent rigid manifold QD fittings **502** (and the effluent rigid manifold QD fittings **412**) going through the rear wall **300**.

[0041] In some implementations, the influent rigid manifold **500** and the effluent rigid manifold **410** may be formed as portions of a single structure (e.g., not separate components). For example, a single rigid manifold may extend across the canister **104** and provide for the influent and effluent fluid connections for the servers.

Example Canister Mounting

[0042] FIG. **6** illustrates an example of the canister **104** and its attachment to the server rack frame **106**. The canister **104** includes the influent rigid manifold **500** with the influent rigid manifold QD fittings **502** attached thereto and the effluent rigid manifold **410** with the effluent rigid manifold QD fittings **412** attached thereto. The influent rigid manifold QD fittings **502** and the effluent rigid manifold QD fittings **412** may be oriented towards a front of the canister **104** and be on opposite sides of the canister **104**. Furthermore, the influent rigid manifold QD fittings **502** and the effluent rigid manifold QD fittings **412** may be arranged in respective linear arrays that are parallel to one another. To gain access to the fluid connections, a top of the canister **104** may have a multi-piece or hinged lid (not shown). In some implementations, the top of the canister **104** may be easily removed for access. The top of the canister **104** has been removed from the illustration to better shown internal components.

[0043] Adjacent to the rigid manifolds may be electrical connections **602** (e.g., one or more per server slot). The electrical connections **602** may provide power and/or communication for the servers within the canister **104**. The electrical connections **602** may connect to the server rack frame **106** at a rear of the canister **104**.

[0044] To support the servers, the canister **104** may contain server portions **604**. The server portions **604** (e.g., brackets, shelves, guides, or rollers) are configured to maintain spacing between the servers within the canister **104** and to ensure proper fluid connections with the QD fittings of the canister **104**.

[0045] The canister **104** is configured to interface with canister portions **606** on the server rack frame **106**. The canister portions **606** (e.g., brackets, shelves, guides, or rollers) are configured to support the canister **104** and couple the canister **104** to the server rack frame **106**. The canister portions **606** may enable the canister **104** to slide into a front of the server rack frame **106** and may have fixing portions (e.g., screws, threaded holes, or latches) configured to secure the canister **104** to the server rack frame **106**.

Example Canister Mounting

[0046] FIG. **7** illustrates an example of a server **700** in relation to the canister **104** while the canister **104** is mounted to the server rack frame **106**. The canister **104** may be mounted to the server rack frame **106** prior to having servers **700** installed therein, or the servers **700** may be installed within the canister **104** and the canister **104** with the servers **700** may be installed within the server rack frame **106**.

[0047] As discussed above, the server **700** may be guided within the canister **104** via the server portions **604**. The server **700** may be inserted such that electrical connections are made with the electrical connections **602** (if implemented), an influent server QD fitting **702** connects with one of the influent rigid manifold QD fittings **502**, and an effluent server QD fitting **704** connects with one of the effluent rigid manifold QD fittings **412**. The QD fittings may be corresponding blind mate fittings (e.g., male on the server **700** and female on the canister **104** or vice-versa). The server **700** may be installed in a front to rear direction and may be secured within the canister **104** using known techniques (e.g., screws, latches, or cams).

[0048] In the present disclosure, the terms, “influent” and “effluent,” refer broadly to the flow of a fluid (e.g., the cooling fluid) and corresponding structures. For example, the term “influent” may imply the flow of fluid moving, or arranged to move, into a manifold, pump, line, or other structure, and the term “effluent” may imply the flow of fluid moving, or arranged to move, out of a manifold, pump, line, or other structure. The fluid moving into the structures (e.g., “influent”) may be the same fluid or a different fluid than that moving out of the structures. The terms, “influent” and “effluent,” in some cases, imply the flow of fluid moving, or arranged to move, in a particular direction (e.g., up, down, right, left, forwards, backwards, or the like). Furthermore, these terms may imply the flow of liquid through or across a boundary (e.g., the cooling fluid moving through, into, or out of a pump, manifold, a chamber, a canister, a reservoir, or the like). Structures such as hoses, pipes, manifolds, quick disconnects, blind connectors, pressure fittings, compression fittings, and others facilitate the influent and effluent movement or flows of fluids, as further described in the present disclosure.

EXAMPLES

[0049] Example 1: A canister for coupling a plurality of servers to a server rack, the canister comprising: an influent rigid manifold including a plurality of influent rigid manifold quick-disconnect (QD) fittings arranged in a fixed orientation and configured to couple with influent server QD fittings on the servers; an effluent rigid manifold including a plurality of effluent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with effluent server QD fittings on the servers; an influent canister QD fitting coupled with the influent rigid manifold and configured to couple with an influent soft manifold QD fitting of an influent soft manifold of the server rack; and an effluent canister QD fitting coupled with the effluent rigid manifold and configured to couple with an effluent soft manifold QD fitting of an effluent soft manifold of the server rack.

[0050] Example 2: The canister of example 1, wherein the influent rigid manifold and the effluent rigid manifold are formed as portions of a single structure.

[0051] Example 3: The canister of example 1, wherein the influent rigid manifold QD fittings and the effluent rigid manifold QD fittings are arranged in respective linear arrays.

[0052] Example 4: The canister of example 3, wherein the linear arrays are parallel.

[0053] Example 5: The canister of example 1, wherein the influent rigid manifold QD fittings and the effluent rigid manifold QD fittings are blind mate fittings.

[0054] Example 6: The canister of example 1, wherein: the canister includes a canister frame; and the influent rigid manifold and the effluent rigid manifold are coupled with the canister frame.

[0055] Example 7: The canister of example 6, wherein the canister frame is formed of sheet metal.

[0056] Example 8: The canister of example 6, wherein the canister frame is configured to mount to the server rack.

[0057] Example 9: The canister of example 6, wherein the canister frame includes server portions configured to orient the servers within the canister.

[0058] Example 10: The canister of example 6, wherein: the canister frame includes a rear wall; the influent rigid manifold and the effluent rigid manifold are mounted on an interior side of the rear wall; and the influent canister QD fitting and the effluent canister QD fitting are disposed on an exterior side of the rear wall.

[0059] Example 11: The canister of example 6, wherein: the canister frame includes a rear wall; the canister is configured to mount the servers on an interior side of the rear wall; the influent rigid manifold and the effluent rigid manifold are mounted on an exterior side of the rear wall; and the influent rigid manifold QD fittings and the effluent rigid manifold QD fittings go through the rear wall.

[0060] Example 12: The canister of example 1, wherein at least one of: the influent canister QD fitting is coupled to the influent rigid manifold via an influent canister hose; or the effluent canister QD fitting is coupled to the effluent rigid manifold via an effluent canister hose.

[0061] Example 13: The canister of example 1 wherein at least one of: the influent canister QD fitting is coupled to the canister via an influent canister QD bracket; or the effluent canister QD fitting is coupled to the canister via an effluent canister QD bracket.

[0062] Example 14: The canister of example 1, wherein the canister is configured to couple three or more servers to the server rack.

[0063] Example 15: The canister of example 1, wherein the canister is configured to receive 19-inch or 21-inch servers.

[0064] Example 16: A cooling system for a server rack, the cooling system comprising: an influent soft manifold including: a plurality of influent hoses; and a plurality of influent soft manifold quick-disconnect (QD) fittings at respective ends of the influent hoses; an effluent soft manifold including: a plurality of effluent hoses; and a plurality of effluent soft manifold QD fittings at respective ends of the effluent hoses; and a plurality of canisters configured to couple respective pluralities of servers to the server rack, each canister including: an influent rigid manifold including a plurality of influent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with influent server QD fittings on the servers; an effluent rigid manifold including a plurality of effluent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with effluent server QD fittings on the servers; an influent canister QD fitting coupled with the influent rigid manifold and configured to couple with one of the influent soft manifold QD fittings of the influent soft manifold; and an effluent canister QD fitting coupled with the effluent rigid manifold and configured to couple with one of the effluent soft manifold QD fittings of the effluent soft manifold.

[0065] Example 17: The cooling system of example 16, wherein the influent rigid manifold QD fittings and the effluent rigid manifold QD fittings are arranged in respective linear arrays.

[0066] Example 18: The cooling system of example 16, wherein each canister includes a canister frame configured to mount to the server rack.

[0067] Example 19: The cooling system of example 16, wherein the influent soft manifold and the effluent soft manifold are coupled with a soft manifold frame.

[0068] Example 20: A server rack comprising: a server rack frame; an influent soft manifold including: a plurality of influent hoses; and a plurality of influent soft manifold quick-disconnect (QD) fittings at respective ends of the influent hoses; an effluent soft manifold including: a plurality of effluent hoses; and a plurality of effluent soft manifold QD fittings at respective ends of the effluent hoses; and a plurality of canisters configured to be mounted to the server rack frame, each canister configured to couple a plurality of servers to the server rack and including: an influent rigid manifold including a plurality of influent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with influent server QD fittings on the servers; an effluent rigid manifold including a plurality of effluent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with effluent server QD fittings on the servers; an influent canister QD fitting coupled with the influent rigid manifold and configured to couple with one of the influent soft manifold QD fittings of the influent soft manifold; and an effluent canister QD fitting coupled with the effluent rigid manifold and configured to couple with one of the effluent soft manifold QD fittings of the effluent soft manifold.

Conclusion

[0069] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of this disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes”, “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Further, the terms up, upper, down, lower, above, below, left, right, forward, rearward, and the like are intended to be understood in the context of the representations described and illustrated above so that a wearable device may have such an orientation in reference to the frame or to various elements as supported by the frame or as illustrated in the drawing figures.

[0070] The corresponding structures, materials, acts, and equivalents of all means or step plus function elements, if any, in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present disclosure has been presented for purposes of illustration and description but is not intended to be exhaustive or limited to this disclosure in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of this disclosure. The various embodiments were chosen and described in order to best explain the principles of this disclosure and the practical application, and to enable others of ordinary skill in the art to understand this disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A canister for coupling a plurality of servers to a server rack, the canister comprising: an influent rigid manifold including a plurality of influent rigid manifold quick-disconnect (QD) fittings arranged in a fixed orientation and configured to couple with influent server QD fittings on the servers; an effluent rigid manifold including a plurality of effluent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with effluent server QD fittings on the servers; an influent canister QD fitting coupled with the influent rigid manifold and configured to couple with an influent soft manifold QD fitting of an influent soft manifold of the server rack; and an effluent canister QD fitting coupled with the effluent rigid manifold and configured to couple with an effluent soft manifold QD fitting of an effluent soft manifold of the server rack.
2. The canister of claim 1, wherein the influent rigid manifold and the effluent rigid manifold are formed as portions of a single structure.
3. The canister of claim 1, wherein the influent rigid manifold QD fittings and the effluent rigid manifold QD fittings are arranged in respective linear arrays.
4. The canister of claim 3, wherein the linear arrays are parallel.
5. The canister of claim 1, wherein the influent rigid manifold QD fittings and the effluent rigid manifold QD fittings are blind mate fittings.
6. The canister of claim 1, wherein: the canister includes a canister frame; and the influent rigid manifold and the effluent rigid manifold are coupled with the canister frame.
7. The canister of claim 6, wherein the canister frame is formed of sheet metal.
8. The canister of claim 6, wherein the canister frame is configured to mount to the server rack.
9. The canister of claim 6, wherein the canister frame includes server portions configured to orient the servers within the canister.
10. The canister of claim 6, wherein: the canister frame includes a rear wall; the influent rigid manifold and the effluent rigid manifold are mounted on an interior side of the rear wall; and the influent canister QD fitting and the effluent canister QD fitting are disposed on an exterior side of the rear wall.
11. The canister of claim 6, wherein: the canister frame includes a rear wall; the canister is configured to mount the servers on an interior side of the rear wall; the influent rigid manifold and the effluent rigid manifold are mounted on an exterior side of the rear wall; and the influent rigid manifold QD fittings and the effluent rigid manifold QD fittings go through the rear wall.
12. The canister of claim 1, wherein at least one of: the influent canister QD fitting is coupled to the influent rigid manifold via an influent canister hose; or the effluent canister QD fitting is coupled to the effluent rigid manifold via an effluent canister hose.
13. The canister of claim 1 wherein at least one of: the influent canister QD fitting is coupled to the canister via an influent canister QD bracket; or the effluent canister QD fitting is coupled to the canister via an effluent canister QD bracket.
14. The canister of claim 1, wherein the canister is configured to couple three or more servers to the server rack.
15. The canister of claim 1, wherein the canister is configured to receive 19 inch or 21 inch servers.
16. A cooling system for a server rack, the cooling system comprising: an influent soft manifold including: a plurality of influent hoses; and a plurality of influent soft manifold quick-disconnect (QD) fittings at respective ends of the influent hoses; an effluent soft manifold including: a plurality of effluent hoses; and a plurality of effluent soft manifold QD fittings at respective ends of the effluent hoses; and a plurality of canisters configured to couple respective pluralities of servers

to the server rack, each canister including: an influent rigid manifold including a plurality of influent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with influent server QD fittings on the servers; an effluent rigid manifold including a plurality of effluent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with effluent server QD fittings on the servers; an influent canister QD fitting coupled with the influent rigid manifold and configured to couple with one of the influent soft manifold QD fittings of the influent soft manifold; and an effluent canister QD fitting coupled with the effluent rigid manifold and configured to couple with one of the effluent soft manifold QD fittings of the effluent soft manifold.

17. The cooling system of claim 16, wherein the influent rigid manifold QD fittings and the effluent rigid manifold QD fittings are arranged in respective linear arrays.

18. The cooling system of claim 16, wherein each canister includes a canister frame configured to mount to the server rack.

19. The cooling system of claim 16, wherein the influent soft manifold and the effluent soft manifold are coupled with a soft manifold frame.

20. A server rack comprising: a server rack frame; an influent soft manifold including: a plurality of influent hoses; and a plurality of influent soft manifold quick-disconnect (QD) fittings at respective ends of the influent hoses; an effluent soft manifold including: a plurality of effluent hoses; and a plurality of effluent soft manifold QD fittings at respective ends of the effluent hoses; and a plurality of canisters configured to be mounted to the server rack frame, each canister configured to couple a plurality of servers to the server rack and including: an influent rigid manifold including a plurality of influent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with influent server QD fittings on the servers; an effluent rigid manifold including a plurality of effluent rigid manifold QD fittings arranged in a fixed orientation and configured to couple with effluent server QD fittings on the servers; an influent canister QD fitting coupled with the influent rigid manifold and configured to couple with one of the influent soft manifold QD fittings of the influent soft manifold; and an effluent canister QD fitting coupled with the effluent rigid manifold and configured to couple with one of the effluent soft manifold QD fittings of the effluent soft manifold.